

FCIM — Feasibility Constraint Identification Module

Architecture Ecosystem: Structured Reality Standards™

Architecture Family: PGA™ — Prediction Governance Architecture

Parent Standard:

Prediction Feasibility Standard (PFES)

Operational Layer: Prediction Feasibility Governance Layer

Category:

Governance & Enforcement

Subcategory:

Prediction Governance

Type: Parent Standard Module

Governed Space:

Feasibility Constraint Identification

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Compatibility: OOF® Methodology OS™ · GOA™ · ORA™ · VALIDOS™ · INTEGROS® · ADIT® · AGA™ · AIG® · ASGA™ · CLIA® · SIMULOS®

AI-Readable: Yes

Authority: OOF®

Protection:

MIP® — Methodological Intellectual Property

Canonical Language:

English (UCL™)

Governed Space

Feasibility Constraint Identification

Canonical Definition

Feasibility Constraint Identification Module (FCIM) defines the governance conditions under which constraints capable of materially enabling, limiting, conditioning, delaying, restricting, or preventing achievement of a predicted outcome are systematically identified, classified, scoped, associated with the Governed Prediction, and preserved for subsequent feasibility analysis.

FCIM establishes the governed constraint landscape surrounding a predicted outcome.



Identifies what may stand between:

the predicted future

and:

the operational ability to achieve that future.

It transforms:

Governed Prediction + Prediction Horizon + Predictive Boundary + Relevant Prediction Basis + Predictive Assumptions + Applicable Operational Conditions

into:

Governed Feasibility Constraint Set

Its foundational principle is:

A predicted outcome cannot be meaningfully assessed for feasibility while material constraints remain unidentified.

Operational Role

FCIM is the first internal governance space of the Prediction Feasibility Standard (PFES) .

Its role is not to determine final feasibility.

Its role is to establish the constraints that the feasibility determination must consider.

The focused operational question is:

What conditions, limits, dependencies, capacities, permissions, resources, or operational realities could materially restrict or prevent achievement of the predicted outcome?

The primary output is:

Governed Feasibility Constraint Set

Internal Module Flow

Feasibility Constraint Identification → Predictive Constraint Mapping → Operational Pathway Assessment
→ Execution & Outcome Feasibility → Feasibility State Determination

Current Position: Feasibility Constraint Identification

Feasibility Constraint

A Feasibility Constraint is a condition, limitation, dependency, boundary, requirement, capacity restriction, permission requirement, or other factor capable of materially affecting whether the predicted outcome can realistically be achieved.

A constraint does not need to make the outcome impossible.

It may:

prevent; restrict; delay; condition; reduce; redirect; scale-limit; sequence-limit; geographically limit; temporally limit

the path to the predicted outcome.

Constraint Identification

Constraint identification establishes which material constraints are relevant to the predicted outcome.

The objective is not to create an unlimited inventory of every conceivable obstacle.

The objective is to identify constraints capable of materially affecting feasibility.

Material Constraint

A constraint is material where its presence, absence, state, threshold, or change could materially alter whether the predicted outcome can be achieved.

Constraint Exists $\neq$ Material Constraint

Constraint Relevance

Every identified constraint should have a defensible relationship to the predicted outcome.

Irrelevant constraints should not inflate the feasibility analysis.

Predicted Outcome Reference

Constraint identification must begin from a clearly defined predicted outcome.

Without an identifiable outcome, constraint relevance cannot be reliably determined.

Prediction Horizon

Constraints are horizon-dependent.

A constraint that prevents achievement within six months may cease to be limiting within five years.

Therefore:

Constraint at Horizon A $\neq$ Constraint at Horizon B

Predictive Boundary

Constraint identification must respect the applicable Predictive Boundary.

Relevant boundaries may include:

geography; jurisdiction; organization; system; population; market; infrastructure; production environment; operational domain.

Internal Constraint

An Internal Constraint originates primarily within the actor, organization, system, or operational environment responsible for achieving the predicted outcome.

Examples include:

insufficient capacity; insufficient personnel; system limitations; internal approval; budget allocation; technical capability.

External Constraint

An External Constraint originates primarily outside the direct control of the actor or system.

Examples include:

regulatory approval; supplier availability; market access; external infrastructure; weather; counterparty action.

Internal $\neq$ Controllable by Default

An internal constraint may remain difficult or impossible to change.

External $\neq$ Unmanageable by Default

An external constraint may be mitigated, substituted, insured, contracted, or otherwise managed.

Hard Constraint

A Hard Constraint prevents the predicted outcome under the applicable conditions unless the constraint itself changes or is removed.

Examples may include:

physical impossibility; mandatory legal prohibition; unavailable indispensable infrastructure; insufficient minimum capacity; incompatible technical limit.

Soft Constraint

A Soft Constraint materially restricts achievement without necessarily preventing it.

Examples may include:

elevated cost; staffing pressure; limited capacity; difficult logistics; implementation complexity.

Hard Constraint ≠ Important Constraint

A constraint may be extremely important without being absolute.

The classification should reflect its operational effect rather than rhetorical severity.

Current Constraint

A Current Constraint exists under the presently applicable conditions.

Potential Constraint

A Potential Constraint may become relevant if specified conditions arise.

Emerging Constraint

An Emerging Constraint is developing and may become material during the Prediction Horizon.

Latent Constraint

A Latent Constraint exists but may remain inactive until another event, threshold, dependency, or condition activates it.

Conditional Constraint

A Conditional Constraint applies only under defined circumstances.

Constraint Activation

A constraint may become active when a triggering condition occurs.

Constraint Trigger

A Constraint Trigger is an event, threshold, state change, or condition capable of activating or materially changing a constraint.

Physical Constraint

Physical constraints arise from physical limits or conditions.

Examples include:

space; mass; temperature; distance; material properties; physical throughput.

Technological Constraint

Technological constraints arise from the capability or limitations of available technology.

Technical Availability ≠ Operational Availability

Technology may exist without being deployable in the required environment.

Capacity Constraint

A Capacity Constraint exists where required throughput, volume, scale, or processing capability exceeds available or realistically obtainable capacity.

Available Capacity

Available Capacity represents capacity that can actually be used within the applicable horizon and conditions.

Planned Capacity

Planned Capacity represents future capacity expected or intended to become available.

Planned Capacity ≠ Available Capacity

Resource Constraint

A Resource Constraint arises where a required resource is insufficient, inaccessible, unavailable, or cannot be obtained within the necessary horizon.

Relevant resources may include:

capital;

materials; energy; compute; data; equipment; land; personnel; specialized expertise.

Resource Existence ≠ Resource Accessibility

Temporal Constraint

A Temporal Constraint arises where the available time is insufficient for required activities, dependencies, approvals, transitions, or execution.

Time Available ≠ Time Required

Sequence Constraint

A Sequence Constraint exists where required actions must occur in a defined order.

Example:

Authorization → Procurement → Construction → Testing → Deployment

Parallel Execution Constraint

Some activities may occur simultaneously.

Others cannot.

Incorrectly assuming parallel execution may create false feasibility.

Dependency Constraint

A Dependency Constraint exists where achievement depends upon another actor, system, event, resource, approval, or condition.

Critical Dependency

A Critical Dependency is one whose failure materially prevents achievement of the predicted outcome.

Dependency Exists ≠ Dependency Will Be Satisfied

Authority Constraint

An Authority Constraint exists where an actor or system lacks required authority to perform an action necessary for the predicted outcome.

Capability ≠ Authority

Permission Constraint

A Permission Constraint may arise from required:

authorization; consent; license; certification; access right; contractual permission; governance approval.

Regulatory Constraint

A Regulatory Constraint arises where applicable regulatory requirements materially affect whether or when the outcome can be achieved.

FCIM identifies the constraint.

It does not determine law.

Legal Constraint

A legal requirement may materially constrain the available operational pathway.

Legal interpretation remains outside FCIM unless supplied through an appropriate governed source.

Financial Constraint

A Financial Constraint exists where capital, liquidity, financing, credit, budget, or other financial capability materially limits achievement.

Economic Constraint

An Economic Constraint exists where the economic conditions required to achieve the outcome are materially incompatible with the prediction.

Economic Constraint ≠ Lack of Commercial Attractiveness by Default

Infrastructure Constraint

Infrastructure constraints may involve:

energy; transportation; communications; cloud infrastructure; production facilities; storage; logistics; physical networks.

Human Capability Constraint

A Human Capability Constraint exists where the predicted outcome requires skills, expertise, availability, training, or coordination that are insufficient or unavailable.

Behavioral Constraint

A Behavioral Constraint exists where achievement materially depends upon human or organizational behavior that cannot reasonably be assumed to occur automatically.

Organizational Constraint

Organizational constraints may include:

governance structure; decision latency; internal approvals; coordination capacity; organizational boundaries; operational responsibilities.

Environmental Constraint

Environmental constraints may arise from:

weather; climate; terrain; environmental limits; natural-resource conditions.

Geographic Constraint

A predicted outcome may be achievable in one location but not another.

Jurisdictional Constraint

Different jurisdictions may impose materially different permissions, requirements, timelines, or operational limitations.

Data Constraint

For predictive or digital systems, achievement may require data with sufficient:

availability; access; coverage; timeliness; compatibility.

FCIM identifies data-related feasibility constraints without replacing PBAS or ADIT®.

Compute Constraint

AI and computational systems may depend upon sufficient:

compute; memory; storage; bandwidth; latency; energy.

Interoperability Constraint

A predicted outcome may depend upon systems that cannot currently exchange information, execute compatible processes, or coordinate reliably.

Integration Constraint

Technology may exist individually while required integration remains unavailable.

Component Availability $\neq$ Integrated System Feasibility

Scale Constraint

An outcome may work at one scale and fail at another.

Small-Scale Capability $\neq$ Large-Scale Feasibility

Throughput Constraint

Required throughput may exceed operational capability even where total capacity appears sufficient.

Bottleneck Constraint

A Bottleneck Constraint limits the overall system despite sufficient capacity elsewhere.

Constraint Concentration

Several feasibility requirements may depend upon one shared constraint.

Example:

production, logistics, and deployment may all depend upon one energy supply.

Number of Feasibility Requirements $\neq$ Number of Independent Constraints

Constraint Dependency

Constraints themselves may depend upon other constraints.

Example:

Capacity expansion may depend upon financing.

Financing may depend upon regulatory approval.

Constraint Chain

A sequence may therefore exist:

Regulatory Approval $\rightarrow$ Financing $\rightarrow$ Capacity Expansion $\rightarrow$ Production Feasibility

Constraint Interaction

Two constraints may interact to create an effect greater than either would create individually.

Constraint Combination

Several soft constraints may collectively create an infeasible environment.

Individually Manageable Constraints $\neq$ Collectively Feasible Conditions

Constraint Conflict

Different sources may disagree about whether a constraint exists or how severe it is.

The disagreement should remain attributable.

Constraint Uncertainty

A constraint may be known while its future state remains uncertain.

Unknown Constraint State $\neq$ Constraint Absent

Constraint Severity

Where useful, constraints may be characterized according to their potential effect.

Possible states include:

Minor

Material

Critical

Outcome-Blocking

Severity should not substitute for the underlying constraint definition.

Constraint Threshold

Some constraints become material only beyond a threshold.

Example:

Production remains feasible up to:

1.2 million units

but the prediction requires:

1.5 million units.

Threshold Breach

A threshold breach may change the constraint state.

Constraint Margin

Where measurable, the difference between required capability and limiting constraint may be preserved.

Positive Margin

Available capability exceeds the requirement.

Negative Margin

The requirement exceeds available capability.

Zero / Near-Zero Margin

The outcome may technically remain feasible while being highly fragile.

Constraint Substitutability

Some constraints can be addressed through alternatives.

Examples:

alternative supplier; alternative infrastructure; alternative technology; alternative financing.

Non-Substitutable Constraint

A Non-Substitutable Constraint has no sufficiently viable alternative within the applicable horizon and boundary.

Constraint Mitigation

A constraint may be mitigated without being removed.

FCIM may preserve identified mitigation pathways but does not treat planned mitigation as already achieved.

Planned Mitigation $\neq$ Resolved Constraint

Constraint Resolution

A constraint is resolved only when the limiting condition no longer materially restricts the applicable predicted outcome.

Constraint Removal $\neq$ Overall Feasibility

Removing one constraint does not establish that all other feasibility conditions are satisfied.

Constraint Completeness

Constraint identification should be sufficiently comprehensive to support subsequent feasibility analysis.

Absolute completeness may be impossible.

The objective is material completeness.

Material Completeness

Material Completeness exists where no reasonably identifiable omitted constraint is known to be capable of materially changing the feasibility determination.

No Identified Constraint ≠ No Constraint Exists

Constraint Identification Confidence

Where material constraints remain uncertain or incompletely identified, the limitation should remain explicit.

Constraint Identification State

Constraint identification may be:

Sufficient

Conditionally Sufficient

Incomplete

Conflicting

Insufficient

Unable to Determine

Constraint Register

Identified material constraints should be preserved in a governed Constraint Register or equivalent machine-readable structure.

Historical Constraint State

Constraint states should remain reconstructable over the prediction lifecycle.

Current Constraint ≠ Historical Constraint

Later Constraint Resolution ≠ Historical Constraint Absence

A constraint resolved later must not be erased from the historical feasibility record.

Constraint Change

Constraints may:

emerge; intensify; weaken; activate; expire; be mitigated; be resolved.

Such changes may materially affect feasibility.

Constraint Change ≠ Prediction Revision by Default

PFES may identify the changed feasibility condition.

PMRS owns prediction-level revision decisions.

Relationship to Prediction Basis

PBAS governs Prediction Basis.

FCIM uses relevant governed information to identify constraints but does not determine the legitimacy of Prediction Basis itself.

Relationship to Predictive Assumptions

PAS governs Predictive Assumptions.

An assumption may concern the expected state of a feasibility constraint.

Example:

Assumption: regulatory approval will be obtained by June.

The approval requirement remains a feasibility constraint even though its future satisfaction is represented as an assumption.

Assumption About Constraint $\neq$ Constraint Itself

Relationship to Probability & Uncertainty

PPUS governs probability and uncertainty.

FCIM does not determine how likely a constraint is to occur.

It identifies whether the constraint is materially relevant to feasibility.

Low Probability Constraint $\neq$ Irrelevant Constraint

Relationship to Operational Reality

ORA™ governs Operational Reality.

FCIM may use governed Operational Reality information to identify current constraints.

It does not independently establish Operational Reality.

Relationship to Evidence

ADIT® governs evidence.

FCIM may reference evidence supporting the existence or state of a constraint without governing evidence validity itself.

Relationship to Feasibility Determination

FCIM does not determine the final feasibility state.

Its governed output establishes the constraint landscape required for subsequent feasibility governance.

Constraint Identification $\neq$ Feasibility Determination

Governance Requirements

FCIM requires that:

1. constraint identification remain associated with a specific Governed Prediction;
2. the predicted outcome remain identifiable;
3. Prediction Horizon remain explicit;
4. Predictive Boundary remain identifiable;
5. materially relevant constraints be systematically identified;
6. irrelevant constraints not be treated as material by default;
7. internal and external constraints remain distinguishable where useful;
8. hard and soft constraints remain distinguishable;
9. current, potential, emerging, latent, and conditional constraints remain representable;
10. constraint triggers remain identifiable where material;
11. physical constraints remain representable;
12. technological constraints remain representable;
13. capacity constraints remain identifiable;
14. planned capacity remain distinguishable from available capacity;
15. resource existence remain distinguishable from resource accessibility;
16. temporal constraints remain identifiable;
17. sequence constraints remain identifiable;
18. critical dependencies remain identifiable;
19. authority and permission constraints remain representable;
20. regulatory and legal constraints remain representable without FCIM independently determining law;
21. financial and economic constraints remain distinguishable where material;

22. infrastructure constraints remain identifiable; 23. human and behavioral constraints remain representable; 24. data, compute, interoperability, and integration constraints remain identifiable where applicable; 25. scale and throughput constraints remain representable; 26. bottleneck constraints remain identifiable; 27. constraint dependencies and chains remain traceable where material; 28. interacting constraints remain assessable; 29. constraint uncertainty remain visible; 30. unknown constraint state not be treated as constraint absence; 31. material thresholds remain identifiable; 32. constraint margins remain representable where measurable; 33. substitutability remain distinguishable from constraint removal; 34. planned mitigation remain distinguishable from resolved constraint; 35. material constraint completeness be assessable; 36. no identified constraint not be treated as proof of unconstrained feasibility; 37. historical constraint states remain reconstructable; 38. later resolution not erase historical constraints; 39. constraint change remain traceable; 40. constraint identification remain distinct from final feasibility determination.

Governance Logic

FCIM applies the following governance logic:

Governed Prediction

Prediction Horizon

Predictive Boundary

Relevant Prediction Basis

Predictive Assumptions

Applicable Operational Conditions

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Predicted Outcome Confirmed

↓

Required Achievement Conditions Considered

↓

Potential Constraint Domains Identified

↓

Material Constraints Identified

↓

Internal / External Constraints Distinguished

↓

Hard / Soft Constraints Distinguished

↓

Current / Potential / Emerging / Conditional Constraints Classified

↓

Capacity / Resource / Temporal / Sequence Constraints Identified

↓

Dependency / Authority / Permission Constraints Identified

↓

Technical / Infrastructure / Human / Economic / Environmental Constraints Identified where applicable

↓

Scale / Integration / Interoperability Constraints Identified where applicable

↓

Constraint Dependencies / Interactions Identified

↓

Constraint Thresholds / Margins Identified where available

↓

Constraint Uncertainty and Conflicts Preserved

↓

Constraint Identification Sufficiency Assessed

↓

Historical Constraint State Preserved

↓

Governed Feasibility Constraint Set

↓

Predictive Constraint Mapping

Where sufficient constraint identification cannot be established:

Feasibility Constraint Identification Gap → Additional Discovery / Clarification / Constraint Qualification /
Restricted Feasibility Analysis / Unable to Determine

Minimum Implementation Framework

A minimum implementation of FCIM must establish:

A. Prediction Reference

The applicable Governed Prediction.

B. Predicted Outcome

The outcome whose feasibility will be assessed.

C. Prediction Horizon

The applicable time horizon.

D. Predictive Boundary

The applicable scope.

E. Constraint Identifier

A distinguishable identifier for each material constraint.

F. Constraint Definition

A clear description of the limiting condition.

G. Constraint Type

At minimum where applicable:

Physical / Technological / Capacity / Resource / Temporal / Sequence / Dependency / Authority /
Permission / Regulatory / Financial / Infrastructure / Human / Behavioral / Environmental / Data / Compute

/ Interoperability / Scale / Other

H. Constraint Classification

At minimum:

Hard / Soft / Unable to Determine

I. Constraint State

At minimum:

Current / Potential / Emerging / Latent / Conditional / Resolved / Unable to Determine

J. Constraint Source

The source from which the constraint was identified.

K. Constraint Scope

The part of the predicted outcome or achievement pathway affected.

L. Dependency / Interaction Reference

Where the constraint depends upon or interacts with another constraint.

M. Threshold / Margin

Where materially measurable.

N. Constraint Uncertainty / Qualification

Any material uncertainty concerning existence, state, severity, or applicability.

O. Historical State

The applicable historical constraint state and version.

Governance Outputs

FCIM may produce:

Governed Feasibility Constraint Set; Feasibility Constraint Register; Constraint Identifier; Constraint Definition; Constraint Type; Constraint Classification; Constraint State; Constraint Source; Constraint Scope; Hard Constraint Record; Soft Constraint Record; Critical Constraint Record; Constraint Trigger; Constraint Dependency Record; Constraint Interaction Record; Constraint Chain; Constraint Threshold; Constraint Margin; Constraint Substitutability Record; Constraint Mitigation Reference; Constraint Conflict Record; Constraint Uncertainty Record; Constraint Identification Sufficiency State; Feasibility Constraint Identification Gap; historical constraint state; Constraint Change Record.

The primary governed output is:

Governed Feasibility Constraint Set

which becomes an input to:

Predictive Constraint Mapping.

Critical Distinctions

A Predicted Outcome Cannot Be Meaningfully Assessed for Feasibility While Material Constraints Remain Unidentified

Constraint Exists $\neq$ Material Constraint

Constraint at Horizon A $\neq$ Constraint at Horizon B

Internal Constraint $\neq$ Controllable by Default

External Constraint ≠ Unmanageable by Default
Hard Constraint ≠ Important Constraint
Technical Availability ≠ Operational Availability
Planned Capacity ≠ Available Capacity
Resource Existence ≠ Resource Accessibility
Time Available ≠ Time Required
Dependency Exists ≠ Dependency Will Be Satisfied
Capability ≠ Authority
Economic Constraint ≠ Lack of Commercial Attractiveness by Default
Component Availability ≠ Integrated System Feasibility
Small-Scale Capability ≠ Large-Scale Feasibility
Number of Feasibility Requirements ≠ Number of Independent Constraints
Individually Manageable Constraints ≠ Collectively Feasible Conditions
Unknown Constraint State ≠ Constraint Absent
Planned Mitigation ≠ Resolved Constraint
Constraint Removal ≠ Overall Feasibility
No Identified Constraint ≠ No Constraint Exists
Current Constraint ≠ Historical Constraint
Later Constraint Resolution ≠ Historical Constraint Absence
Constraint Change ≠ Prediction Revision by Default
Assumption About Constraint ≠ Constraint Itself
Low Probability Constraint ≠ Irrelevant Constraint
Constraint Identification ≠ Feasibility Determination

Enterprise Application

FCIM gives organizations a governed method for identifying what could materially prevent a predicted future from becoming operationally achievable.

It is particularly relevant where predictions depend upon complex combinations of:

capacity; resources; infrastructure; approvals; technology; suppliers; timelines; human capability; system integration; external dependencies.

Instead of asking only:

How likely is the predicted outcome?

FCIM introduces the prior feasibility question:

What could materially prevent this outcome from being achieved?

Use Case 1 — Demand Forecast Exceeds Physical Production Capacity

A manufacturer predicts:

1.6 million units sold next year.

The commercial forecasting system is focused on customer demand and produces a high-probability sales forecast.

The production facility, however, has a maximum current capacity of:

1.15 million units per year.

Management plans a capacity expansion, but the additional production line is not expected to become operational until after the Prediction Horizon.

The critical governance question is:

Which material constraint prevents the predicted sales outcome from being operationally achievable within the stated horizon?

FCIM identifies:

Capacity Constraint

classifies current versus planned capacity, associates the constraint with the Prediction Horizon, and preserves the capacity gap for subsequent feasibility analysis.

Planned Capacity $\neq$ Available Capacity

Use Case 2 — Every Technology Exists, but the System Still Cannot Be Deployed

An organization predicts that a new autonomous logistics system will be fully operational across its network within twelve months.

The required technologies individually exist:

autonomous vehicles; routing software; warehouse automation; cloud infrastructure; sensor networks.

However, the vehicles cannot communicate with the existing warehouse-control system, several facilities lack required network infrastructure, and regulatory authorization for autonomous operation has not yet been obtained.

The critical governance question is:

Can individual technological availability be treated as evidence that the complete operational system is feasible?

FCIM identifies the relevant:

Integration Constraint

Interoperability Constraint

Infrastructure Constraint

and:

Permission Constraint

before the final feasibility state is determined.

Component Availability $\neq$ Integrated System Feasibility

Canonical Boundary

FCIM governs the identification, definition, classification, source, scope, state, threshold, margin, dependency, interaction, uncertainty, substitutability, historical preservation, and material completeness of constraints capable of affecting Prediction Feasibility. It does not determine the final feasibility state, Prediction Probability, Predictive Uncertainty, Prediction Formation, Operational Reality, evidence validity, prediction-level revision, or predictive performance.

FCIM governs:

what may materially limit or prevent achievement of the predicted outcome.

It does not determine:

whether the complete predicted outcome is ultimately feasible.

Canonical Closing Statement

A prediction can appear achievable until the constraints surrounding it are made visible.

FCIM establishes the governed constraint landscape before feasibility is judged, preventing hidden limits, missing dependencies, unavailable capacity, and unresolved permissions from disappearing behind a predictive result.

Feasibility begins by identifying what reality will not allow the prediction to ignore.

[↑](#)

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